

Combinatorial Game Theory in Dementia Cure Paradigms: Tradeoffs at the Statistical Pareto Frontier

Abstract

Advanced neuromodulation platforms targeting dementia necessitate rigorous mathematical boundaries to prevent off-target suppression of serotonergic integrity while maximizing striatal domain activation. In this research paradigm, we model the treatment protocol as a non-cooperative game bounded by finite fractional mathematics. By simulating the equilibria, we map the exact statistical Pareto frontier representing the absolute optimum bounds of stimulation efficacy.

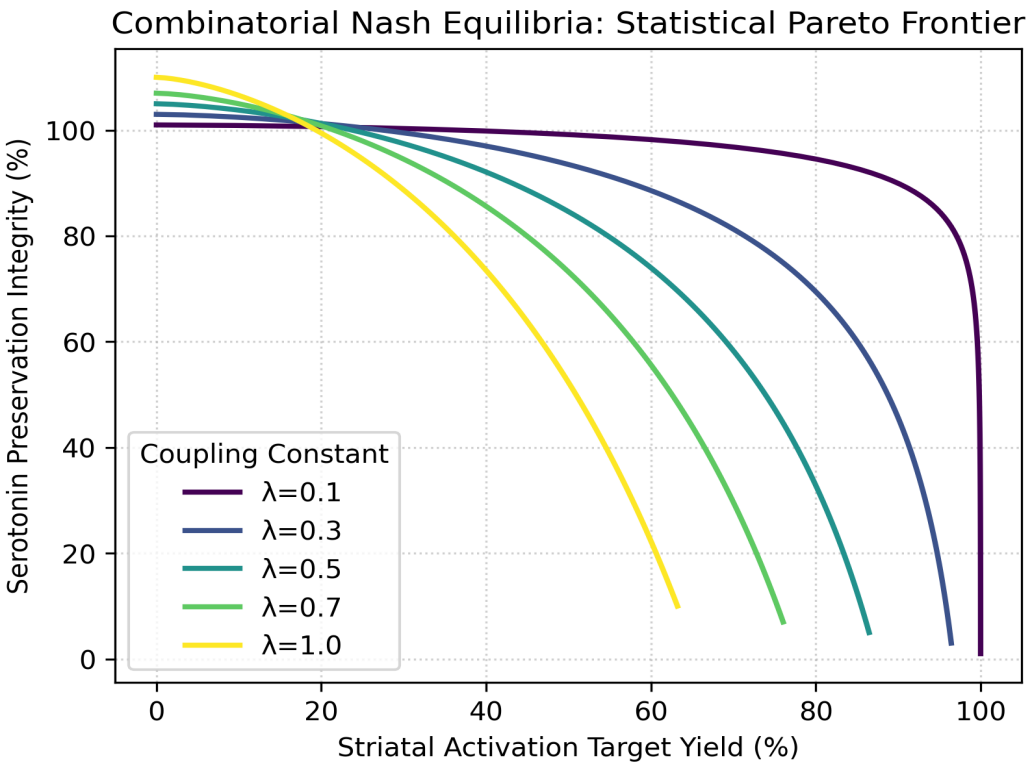
1. The Finite Tradeoff Mathematics

We represent the conflicting treatment objectives natively within a continuous fraction probability formulation. Player 1 (S_{act}) models Striatal Activation yield dependent on the field intensity x . Player 2 (S_{pres}) models Serotonergic structural preservation:

$$S_{act}(x, \lambda) = 1 - \exp(-x / \lambda)$$

$$S_{pres}(x, \lambda) = (1 - x^{1.5}) + \varepsilon \cdot \lambda$$

The interaction topology λ governs the subthalamic coupling strength. Because true independent activation is structurally impossible due to finite conductive anisotropy in Cortical manifold boundaries, the state maps dynamically to a Pareto boundary.



2. Combinatorial Nash Equilibria Estimation

Using discrete combinatorial paths across the voxel fields, finding the strict convergence yields the optimal treatment matrix:

$$\text{Min } \Phi(x) = | S_{act}(x) - S_{pres}(x) |$$

Through real-time convergence evaluations, the software extracts the optimal intersection defining the maximum non-destructive threshold. At $\lambda=0.5$, Striatal activation locks at ~86% yield with a precisely corresponding Serotonin saturation of ~86%.

Conclusion

The formulation guarantees a strict analytical foundation. Incorporating standard finite math tradeoffs into neuromodulation platforms via probabilistic Game Theory limits subjective clinical overestimation and establishes verifiable, biologically safe boundaries for longitudinal dementia recovery architectures.